Adversarial robustness in fraud detection

"How do fraudsters adapt to fraud detection models, and what methods do researchers propose to keep fraud detection robust against changing patterns, such as adversarial robustness or continual learning?" *Generated by LeapSpace*

Fraudsters adapt by **changing the transaction patterns, perturbing inputs, poisoning training data, and hiding inside graph structure**, so researchers increasingly defend fraud detectors with **adversarial training, continual learning, drift detection, robust aggregation, and graph-specific robustness methods** [64] [172] [173] [174] [175] [176] [177] [178] [179].

How fraudsters adapt

Adaptation pattern	What it means in practice	Why it breaks detectors
Evasion / perturbation	Attackers slightly change transaction features or behavior to slip past a classifier, including transferable black-box attacks and simple single-feature changes [173] [174] [176] [180].	Models trained on fixed patterns can be fooled by small input changes, especially when they rely on stable feature correlations [173] [176].
Poisoning / data corruption	Attackers corrupt training data or inject malicious graph structure so future models learn the wrong boundaries [178] [181] [182].	Poisoned training shifts the learned decision rule, which can degrade performance even after retraining [178] [182].
Camouflage / collusion	Fraud gangs hide their behavior inside normal-looking neighborhoods or collusive structures in graphs [172] [179].	Graph detectors can miss coordinated fraud when malicious nodes are embedded in seemingly benign local patterns [172].
Concept drift exploitation	Fraudsters keep changing tactics so the data distribution moves over time [183] [184] [185] [186] [187] [188] [189].	Static models become stale, and their performance degrades as the fraud pattern shifts [183] [185] [188].

Methods researchers propose to stay robust

Defense family	Core idea	Representative methods / findings
Adversarial training	Train on attacked or perturbed examples so the detector learns to resist them [55] [174] [178] [180] [184] [186] [190] [191] [192] [193].	Widely recommended as the main robustness strategy, including for graph-based fraud detectors and dynamic graphs [55] [178] [190].

Defense family	Core idea	Representative methods / findings
Continual / online learning	Update the model over time without forgetting earlier fraud knowledge [185] [186] [187] [188] [194] [195] [196].	Continual learning is repeatedly framed as the answer to evolving fraud patterns and catastrophic forgetting [185] [187].
Drift detection and adaptive updating	Detect when behavior changes, then adapt thresholds, features, or retraining windows [183] [197] [198] [199] [200] [201] [202] [203] [204].	Systems such as DriftShield, FinDriftXplain, CADE, and ADAPT aim to spot drift early and trigger targeted adaptation rather than full retraining [183] [197] [198] [203].
Graph robustness methods	Denoise or reweight suspicious edges and improve graph resilience to manipulation [205] [206] [207] [208] [209] [210].	Survey and model papers recommend graph structure learning, perturbation-aware regularization, and consistency constraints when transaction graphs may be manipulated [205] [206] [210].
Reweighting, replay, and memory	Use replay buffers, dynamic reweighting, or memory modules to retain rare fraud signals while adapting to new ones [183] [195] [196] [207] [211].	DriftShield's adaptive replay buffer and other memory-aware methods are designed to balance rapid adaptation with forgetting prevention [183] [211].
Robust aggregation / federated defenses	In collaborative settings, use more robust update aggregation and privacy-preserving verification [64] [195] [212].	The literature notes median/Krum-style aggregation, independent checks, and secure mechanisms as defenses against malicious or corrupted updates [64] [212].

What seems most effective

- **For changing fraud patterns**, the strongest theme is **continual learning plus drift-aware updating**, because many studies say the main failure mode is not a single attack but a moving target [185] [187] [188] [197].
- **For deliberate adversarial manipulation**, **adversarial training** is the most direct defense, especially when paired with graph consistency or robust feature extraction [178] [184] [190] [191] [192].
- **For graph-based fraud**, researchers increasingly combine **temporal modeling** with **robust graph learning**, because static graph assumptions become stale when fraudsters change behavior [187] [188] [195] [206].

Bottom line

Fraudsters win when detectors stay static; researchers respond by making models **attack-aware, drift-aware, and memory-aware** so the detector can keep learning while preserving older fraud knowledge [183] [185] [187] [188]. The clearest current direction is a combination of **adversarial training for deliberate evasion** and **continual learning for slow behavioral change**, with graph-specific robustness methods for transaction networks [178] [190] [192] [206].

Confidence level: **High** — multiple independent studies across fraud detection, graph learning, and security applications converge on the same adaptation/defense pattern, though exact best methods vary by dataset and deployment setting [64] [183] [185] [187] [188].

References

55. Dasari, B., Dhiraj, T.S., Jambhrunkar, G., (...), Simmhan, Y. (2026) Billion-Scale Fintech Analytics: Scalable Data Management and Anomaly Detection at NPCI. *Proceedings - 2026 IEEE 42nd International Conference on Data Engineering, ICDE 2026*. https://doi.org/10.1109/ICDE65706.2026.00300

64. Tong, G., Shen, J. (2023) Financial transaction fraud detector based on imbalance learning and graph neural network. *Applied Soft Computing*. https://doi.org/10.1016/j.asoc.2023.110984

172. Choi, J., Kim, H., Whang, J.J. (2025) Unveiling the Threat of Fraud Gangs to Graph Neural Networks: Multi-Target Graph Injection Attacks Against GNN-Based Fraud Detectors. *Proceedings of the AAAI Conference on Artificial Intelligence*. https://doi.org/10.1609/aaai.v39i15.33760

173. Miljković, A., Gnjatović, M., Joksimović, M., (...), Čabarkapa, M. (2026) White-Box and Transfer-Based Black-Box Adversarial Attacks on Neural Network Models for Credit Card Fraud Detection. *Electronics (Switzerland)*. https://doi.org/10.3390/electronics15184297

174. Ullah, H., Thahsin Zahir Ismail, A., Babu-Saheer, L., Maktabdar-Oghaz, M. (2022) Credit Card Fraud Using Adversarial Attacks. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-031-21441-7_26

175. Lunghi, D., Molinghen, Y., Simitsis, A., (...), Bontempi, G. (2026) Fraud-RLA: A Reinforcement Learning Adversarial Attack Against Credit Card Fraud Detection. *IEEE Transactions on Dependable and Secure Computing*. https://doi.org/10.1109/TDSC.2026.3673847

176. Fok, J.L., Zeng, Q., Chen, S., (...), Chen, H. (2025) Foe for Fraud: Transferable Adversarial Attacks in Credit Card Fraud Detection. *Proceedings of the IEEE International Conference on Web Services, ICWS*. https://doi.org/10.1109/ICWS67624.2025.00043

177. Xiao, J., Tian, Y., Jia, Y., (...), Wang, S. (2023) Black-Box Attack-Based Security Evaluation Framework for Credit Card Fraud Detection Models. *INFORMS Journal on Computing*. https://doi.org/10.1287/IJOC.2023.1297

178. Zhao, Z., Yang, Y., Yin, Z., (...), Chen, E. (2024) Adversarial Attack and Defense on Discrete Time Dynamic Graphs. *IEEE Transactions on Knowledge and Data Engineering*. https://doi.org/10.1109/TKDE.2024.3438238

179. Lunghi, D., Simitsis, A., Caelen, O., Bontempi, G. (2023) Adversarial Learning in Real-World Fraud Detection: Challenges and Perspectives. *ACM International Conference Proceeding Series*. https://doi.org/10.1145/3600046.3600051

180. Alghureid, A., Mohaiseny, D. (2025) Simple Perturbations Subvert Ethereum Phishing Transactions Detection: An Empirical Analysis. *Lecture Notes in Computer Science*. https://doi.org/10.1007/978-981-96-1624-4_10

181. Bisht, B., Verma, R., Yadav, V., (...), Kaushik, Y. (2025) Adversarial AI for Cybersecurity to Mitigate AI-Induced Attacks. *2025 IEEE 7th International Conference on Computing, Communication and Automation, ICCCA 2025*. https://doi.org/10.1109/ICCCA66364.2025.11325530

182. Pal, H., Bachina, V.S.P., Gangwal, A., Sharma, C. (2026) LORETTA: A Low Resource Framework to Poison Continuous Time Dynamic Graphs. *Proceedings of the AAAI Conference on Artificial Intelligence*. https://doi.org/10.1609/aaai.v40i10.37770

183. Cao, J., Zheng, W., Ge, Y., Wang, J. (2025) DriftShield: Autonomous Fraud Detection via Actor-Critic Reinforcement Learning With Dynamic Feature Reweighting. *IEEE Open Journal of the Computer Society*. https://doi.org/10.1109/OJCS.2025.3587001

184. Crisostomo, Joao, Bacao, Fernando, Lobo, Victor (2026) Machine learning methods for fraud detection within Ethereum blockchain—A review. *Blockchain: Research and Applications*. https://doi.org/10.1016/j.bcra.2026.100469

185. Manurung, H., Arinengsih, D.R. (2026) Robust machine learning for fraud detection under imbalance, rarity, and adversarial behavior: Evidence across financial reporting, payments, and graph ecosystems. *Social Sciences and Humanities Open*. https://doi.org/10.1016/j.ssaho.2026.103697

186. Manivannan, K., Maheswari, R., Shanmugam, S., Suresh Babu, A. (2026) Hybrid deep learning approach with capsule networks for financial fraud detection. *Ain Shams Engineering Journal*. https://doi.org/10.1016/j.asej.2026.104405

187. Deprez, B., Wei, W., Verbeke, W., (...), Verdonck, T. (2025) Advances in Continual Graph Learning for Anti-Money Laundering Systems: A Comprehensive Review. *Wiley Interdisciplinary Reviews: Computational Statistics*. https://doi.org/10.1002/wics.70040

188. Menezes, R.S., Holanda Filho, R.H. (2025) Investigating the Robustness of Graph Neural Networks to Data Drift: A Case Study on Financial Transaction Data. *IEEE Access*. https://doi.org/10.1109/ACCESS.2025.3611817

189. Iwashita, A.S., Papa, J.P. (2019) An Overview on Concept Drift Learning. *IEEE Access*. https://doi.org/10.1109/ACCESS.2018.2886026

190. Kumar, Srijan (2023) Advances in AI for web integrity, equity, and well-being. *Frontiers in Big Data*. https://doi.org/10.3389/fdata.2023.1125083

191. Qian, Z., Huang, K., Wang, Q.-F., Zhang, X.-Y. (2022) A survey of robust adversarial training in pattern recognition: Fundamental, theory, and methodologies. *Pattern Recognition*. https://doi.org/10.1016/j.patcog.2022.108889

192. Qiao, Y., Sathyanarayana, N.B., Shi, C., (...), Hou, T. (2026) A survey on adversarial machine learning: Attacks, defenses, real-world applications, and future research directions. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2026.132670

193. Lazarov, G., Kobbaey, T., Akre, V., (...), Diab, J. (2025) Adversarial Artificial Intelligence (AI) in Cybercrime Detection and Forensics. *2025 10th International Conference on Information Technology Trends, ITT 2025*. https://doi.org/10.1109/ITT69610.2025.11352927

194. https://www.sciencedirect.com/science/article/pii/S2731080926005983

195. https://www.sciencedirect.com/science/article/pii/S1573756X26000733

196. Benatia, M.A., Hafsi, M., Ben Ayed, S. (2025) A continual learning approach for failure prediction under non-stationary conditions: Application to condition monitoring data streams. *Computers and Industrial Engineering*. https://doi.org/10.1016/j.cie.2025.111049

197. Muthusenthil, B., Ragul, M., Sathishkumar, B., Raahuleshwar, A. (2026) FinDriftXplain: Unsupervised Drift Detection and Characterization in Deep Learning Embeddings for Financial Fraud Detection. *Proceedings - IEEE International Conference on Future and Advanced Computing Technologies, ICFACT 2026*. https://doi.org/10.1109/ICFACT66887.2026.11518297

198. Mai, T.-D., Hoang, K., Baigutanova, A., (...), Kim, S. (2021) Customs Fraud Detection in the Presence of Concept Drift. *IEEE International Conference on Data Mining Workshops, ICDMW*. https://doi.org/10.1109/ICDMW53433.2021.00052

199. Malekian, D., Hashemi, M.R. (2013) An adaptive profile based fraud detection framework for handling concept drift. *2013 10th International ISC Conference on Information Security and Cryptology, ISCISC 2013*. https://doi.org/10.1109/ISCISC.2013.6767338

200. Yu, X., Ruan, L., Evans, J.S., Wong, E. (2025) Adaptive windowing-based concept drift detection and adaptation framework for human-to-machine applications over future communication networks. *Journal of Optical Communications and Networking*. https://doi.org/10.1364/JOCN.538964

201. Pereira, E.V., Silva, W.S.D. (2025) A Comparison of Approaches for Handling Concept Drifts in Data Processed With Machine Learning. *IEEE Access*. https://doi.org/10.1109/ACCESS.2025.3557229

202. Thakur, R.K., Kumari, R. (2026) A Hybrid Statistical Framework for Detecting Data Drift and Transactional Fraud in Financial Systems. *2026 11th International Conference on Cloud Computing and Big Data Analytics, ICCCBDA 2026*. https://doi.org/10.1109/ICCCBDA69573.2026.11676520

203. Yang, L., Guo, W., Hao, Q., (...), Wang, G. (2021) CADE: Detecting and explaining concept drift samples for security applications. *Proceedings of the 30th USENIX Security Symposium*. https://www.scopus.com/pages/publications/85112826522

204. Zhao, D., Koh, Y.S. (2020) Feature drift detection in evolving data streams. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-030-59051-2_23

205. https://www.sciencedirect.com/science/article/pii/S2364416826002465

206. Zhou, P., Yin, K., Zhu, H., (...), Chen, Y. (2026) A survey on graph structure learning. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2026.134114

207. Huang, Gang, Zhang, Lu, Wang, Hailun (2025) Improving model adversarial robustness in Extractive Question Answering via Wasserstein-Guided feature Representations. *Alexandria Engineering Journal*. https://doi.org/10.1016/j.aej.2025.05.003

208. Yu, H., Liu, W., Zhu, N., (...), Luo, X. (2024) IN-GFD: An Interpretable Graph Fraud Detection Model for Spam Reviews. *IEEE Transactions on Artificial Intelligence*. https://doi.org/10.1109/TAI.2024.3420262

209. Chikwendu, I.A., Zhang, X., Ukwuoma, C.C., (...), Al-antari, M.A. (2025) Spectrum-Constrained and Skip-Enhanced Graph Fraud Detection: Addressing Heterophily in Fraud Detection with Spectral and Spatial Modeling. *Symmetry*. https://doi.org/10.3390/sym17040476

210. Khan, A., Ke, X., Wu, Y. (2025) Graph Data Management and Graph Machine Learning: Synergies and Opportunities. *SIGMOD Record*. https://doi.org/10.1145/3749116.3749122

211. Kurniawan, M., Putra, H., Mintaraga, C.A., Hidayaturrahman, u. (2025) Sequential Oversampling for Fraud Detection: Leveraging Generative Adversarial Networks and Continual Learning Approach in Imbalanced Data Streams. *Procedia Computer Science*. https://doi.org/10.1016/j.procs.2025.08.301

212. Zhang, Z., Zhang, C., Chen, G., (...), Huang, L. (2026) Distinguishing good from bad in federated learning: A novel data fraud detection method using prototype learning and variational autoencoder. *Decision Support Systems*. https://doi.org/10.1016/j.dss.2026.114703